

Trainings ▾

General Information

The QSK19 uses push rods that are machined from solid bar stock instead of tubing. The ball end and the socket are machined on the rod.

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It is a good service practice, to mark the push rods for location when removing them from the engine. The valve and injector push rods are not interchangeable. The cam follower assembly **must** be replaced if the push rod is worn or damaged in the area where it contacts the socket.

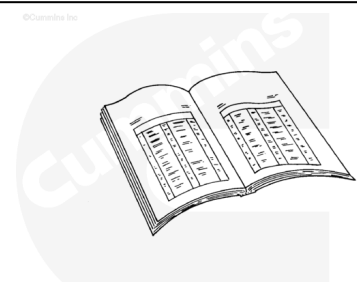
Note : Replace both the rocker lever adjusting screw and the push rod if the socket surface in the rod or on the adjusting screw is damaged.

Preparatory Steps

with Mechanically Actuated Injector

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)

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with Electronically Actuated Injector

- Remove the high pressure fuel lines. Refer to Procedure 006-051 in Section (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)

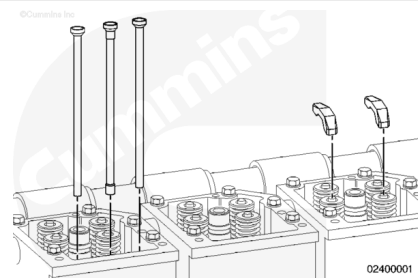
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- Remove the rocker levers. Refer to Procedure 003-009 in Section (/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html)

Remove with Mechanically Actuated Injector

It is a good service practice to mark the push rods so they can be installed in their original position.

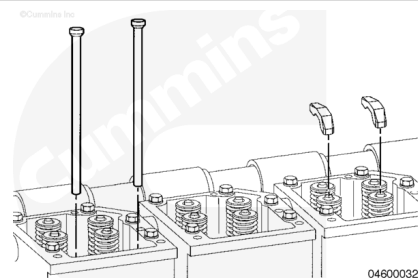
Remove the push rods and crossheads.



with Electronically Actuated Injector

It is recommended a service practice to mark the push rods so they can be installed in their original position.

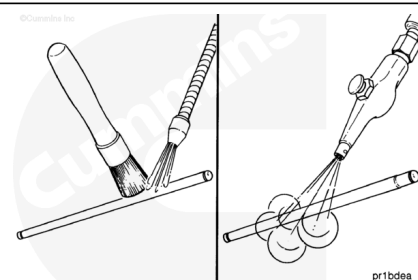
Remove the push rods.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

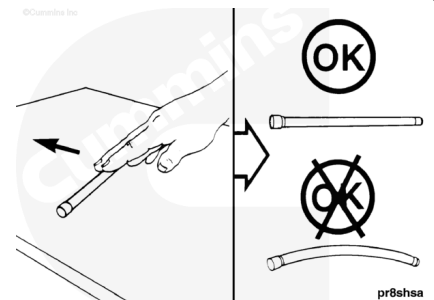
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use solvent to clean the push rods.

Dry the push rods with compressed air.

Clean and inspect the push rods. Inspect to be sure the rod is straight. View the rod while rolling it across a level surface.

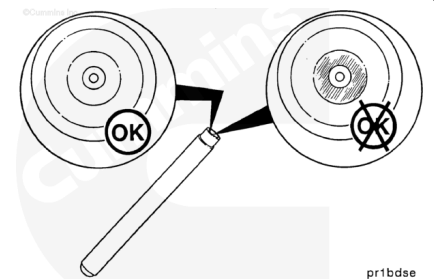
Note : Do not use or try to straighten a bent push rod.



Inspect the socket end of the push rod for uneven wear or scratches.

If a worn socket is found, the mating adjusting screw in the rocker lever **must** also be replaced.

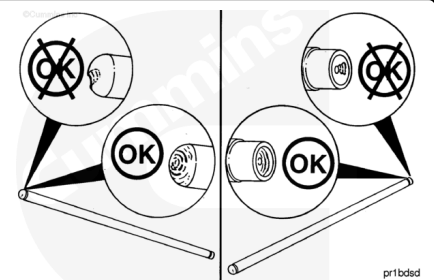
When parallel scratches are found in the contact area, the push rod **must** be replaced.



Inspect both ends of the push rods for wear or damage.

Note : Replace both the rocker lever adjusting screw and the push rod if the socket surface in the rod or on the adjusting screw is damaged.

The cam follower assembly **must** be replaced if the push rod is worn or damaged in the area where it contacts the socket.

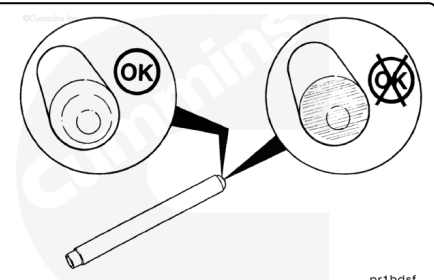


Inspect the ball end of the push rod.

The contact area **must** show a smooth seating pattern.

If the ball end of the push rod has parallel grooves and scratches with a raised center, the push rod **must** be replaced.

Note : If a worn push rod is found, the mating cam assembly socket must be replaced.



Install

with Mechanically Actuated Injector

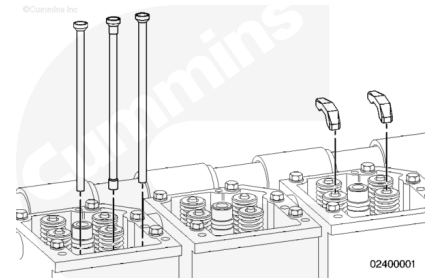
Note : The push rods must be seated correctly in the cam followers.

Note : Mechanically Actuated Injectors: The injector push rods are thicker in the middle. The valve push rods are the same for both intake and exhaust.

Use engine oil to lubricate the sockets in the cam followers.

Install the push rods.

Install the crossheads.



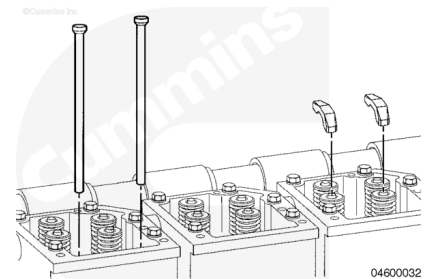
with Electronically Actuated Injector

Note : The push rods must be seated correctly in the cam followers.

Use clean engine oil to lubricate the sockets in the cam followers.

Install the push rods.

Install the crossheads.



Finishing Steps

with Mechanically Actuated Injector

- Install the rocker levers. Refer to Procedure 003-009 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Adjust the overhead (OBC). Refer to Procedure 003-006 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html>)
- Install the rocker lever covers and all related parts. Refer to Procedure 003-011 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Operate the engine and check for leaks.



with Electronically Actuated Injector

- Install the rocker levers. Refer to Procedure 003-009 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Install the rocker lever covers. Refer to Procedure 003-011 in Section (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Install the high pressure fuel lines. Refer to Procedure 006-051 in Section (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Operate the engine and check for leaks.